

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title : Lenskart E-commerce Database Management System

1. Database Design

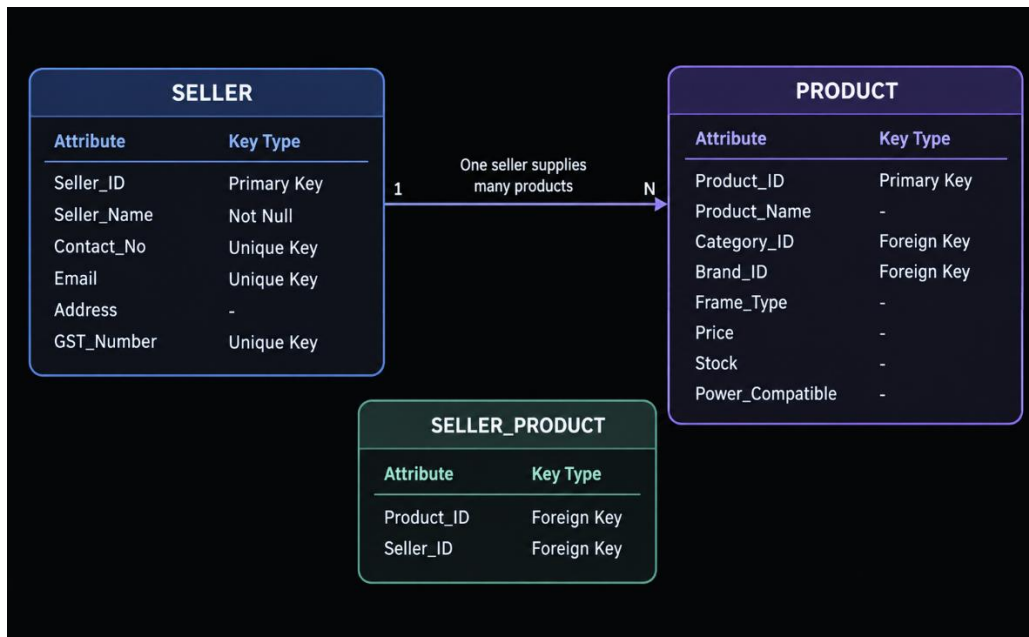
1.1 Seller Table

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	Not Null
Contact_No	Unique Key
Email	Unique Key
Address	-
GST_Number	Unique Key

1.2 Inventory Table

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Store_ID	Foreign Key
Seller_ID	Foreign Key
Quantity_Available	-
Last_Updated	-

2. Table Relationships



3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (
2     Seller_ID NUMBER PRIMARY KEY,
3     Seller_Name VARCHAR2(100) NOT NULL,
4     Contact_No VARCHAR2(15) UNIQUE,
5     Email VARCHAR2(100) UNIQUE,
6     Address VARCHAR2(200),
7     GST_Number VARCHAR2(20) UNIQUE
8 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (
2     Inventory_ID NUMBER PRIMARY KEY,
3     Product_ID NUMBER,
4     Store_ID NUMBER,
5     Seller_ID NUMBER,
6     Quantity_Available NUMBER NOT NULL,
7     Last_Updated DATE,
8     CONSTRAINT fk_inventory_product
9         FOREIGN KEY (Product_ID)
10        REFERENCES Product(Product_ID),
11     CONSTRAINT fk_inventory_store
12        FOREIGN KEY (Store_ID)
13        REFERENCES Store(Store_ID),
14     CONSTRAINT fk_inventory_seller
15        FOREIGN KEY (Seller_ID)
16        REFERENCES Seller(Seller_ID)
17 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller
  2 (Seller_ID, Seller_Name, Contact_No, Email, Address, GST_Number)
  3 VALUES
  4 (101, 'ABC Traders', '9876543210', 'abc@gmail.com',
  5 'Chennai, Tamil Nadu', '33ABCDE1234F1Z5');

1 row created.
```

```
SQL> INSERT INTO Seller
  2 (Seller_ID, Seller_Name, Contact_No, Email, Address, GST_Number)
  3 VALUES
  4 (102, 'Vision Opticals', '9876543211', 'vision@gmail.com',
  5 'Coimbatore, Tamil Nadu', '33FGHIJ5678K2Z6');

1 row created.
```

```
SQL> INSERT INTO Seller
  2 (Seller_ID, Seller_Name, Contact_No, Email, Address, GST_Number)
  3 VALUES
  4 (104, 'Clear Vision', '9876543213', 'clearvision@gmail.com',
  5 'Salem, Tamil Nadu', '33PQRST3456U4Z8');

1 row created.
```

```
SQL> INSERT INTO Seller
  2 (Seller_ID, Seller_Name, Contact_No, Email, Address, GST_Number)
  3 VALUES
  4 (104, 'Clear Vision', '9876543213', 'clearvision@gmail.com',
  5 'Salem, Tamil Nadu', '33PQRST3456U4Z8');

1 row created.
```

```
SQL> INSERT INTO Seller
  2 (Seller_ID, Seller_Name, Contact_No, Email, Address, GST_Number)
  3 VALUES
  4 (105, 'Eye Care Traders', '9876543214', 'eyecare@gmail.com',
  5 'Madurai, Tamil Nadu', '33VWXYZ7890A5Z9');

1 row created.
```

5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory
  2 VALUES (1, 201, 1, 101, 50, TO_DATE('22-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```
SQL> INSERT INTO Inventory
  2 VALUES (2, 202, 1, 102, 30, TO_DATE('22-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```
SQL> INSERT INTO Inventory
  2 VALUES (3, 203, 2, 103, 45, TO_DATE('21-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```
SQL> INSERT INTO Inventory
  2 VALUES (4, 204, 2, 104, 25, TO_DATE('20-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```
SQL> INSERT INTO Inventory
2 VALUES (5, 205, 3, 105, 60, TO_DATE('19-09-2026', 'DD-MM-YYYY'));

1 row created.
```

6. Seller Product Information

6.1 Display Seller and Product Details

This query combines Seller, Inventory, and Product Information.

```
SQL> SELECT
2     s.Seller_Name,
3     p.Product_Name,
4     p.Price,
5     i.Quantity_Available
6 FROM Seller s
7 JOIN Inventory i
8     ON s.Seller_ID = i.Seller_ID
9 JOIN Product p
10    ON i.Product_ID = p.Product_ID;
```

SELLER_NAME

PRODUCT_NAME

PRICE QUANTITY_AVAILABLE

ABC Traders

Classic Eyeglasses

1200

50

Vision Opticals

Stylish Sunglasses

1500

30

SELLER_NAME

PRODUCT_NAME

PRICE QUANTITY_AVAILABLE

Bright Eyewear

Reading Glasses

800

45

Clear Vision

Contact Lenses

SELLER_NAME

PRODUCT_NAME

PRICE QUANTITY_AVAILABLE

1000

25

Eye Care Traders

Blue Light Glasses

1300

60

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT
2     p.Product_ID,
3     p.Product_Name,
4     i.Quantity_Available
5 FROM Product p
6 JOIN Inventory i
7   ON p.Product_ID = i.Product_ID
8 WHERE i.Quantity_Available > 0;
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
QUANTITY_AVAILABLE
```

```
-----
```

```
      201
Classic Eyeglasses
      50
```

```
      202
Stylish Sunglasses
      30
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
QUANTITY_AVAILABLE
```

```
-----
```

```
      203
Reading Glasses
      45
```

```
      204
Contact Lenses
```

```
PRODUCT_ID
```

```
-----
```

```
PRODUCT_NAME
```

```
-----
```

```
QUANTITY_AVAILABLE
```

```
-----
```

```
      25
```

```
      205
Blue Light Glasses
      60
```

7.2 Display Unavailable Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     i.Quantity_Available
  5 FROM Product p
  6 JOIN Inventory i
  7   ON p.Product_ID = i.Product_ID
  8 WHERE i.Quantity_Available = 0;

PRODUCT_ID
-----
PRODUCT_NAME
-----
QUANTITY_AVAILABLE
-----
          205
Blue Light Glasses
          0

```

8. Inventory Update

Demonstrate updating stock information.

```

SQL> UPDATE Inventory
  2 SET Quantity_Available = 20,
  3     Last_Updated = TO_DATE('22-09-2026', 'DD-MM-YYYY')
  4 WHERE Product_ID = 205;

1 row updated.

SQL>
SQL> commit;

Commit complete.

SQL> SELECT
  2     Product_ID,
  3     Quantity_Available,
  4     Last_Updated
  5 FROM Inventory
  6 WHERE Product_ID = 205;

PRODUCT_ID QUANTITY_AVAILABLE LAST_UPDA
-----
          205                20 22-SEP-26

```

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SQL> SELECT
2     i.Inventory_ID,
3     p.Product_ID,
4     p.Product_Name,
5     s.Store_ID,
6     s.Store_Name,
7     se.Seller_ID,
8     se.Seller_Name,
9     i.Quantity_Available,
10    p.Price,
11    i.Last_Updated
12 FROM Inventory i
13 JOIN Product p
14     ON i.Product_ID = p.Product_ID
15 JOIN Store s
16     ON i.Store_ID = s.Store_ID
17 JOIN Seller se
18     ON i.Seller_ID = se.Seller_ID
19 ORDER BY i.Inventory_ID;

```

```

INVENTORY_ID PRODUCT_ID
-----

```

```

PRODUCT_NAME
-----

```

```

STORE_ID
-----

```

```

STORE_NAME
-----

```

```

SELLER_ID
-----

```

```

SELLER_NAME
-----

```

```

-----
QUANTITY_AVAILABLE    PRICE LAST_UPDA
-----

```

```

1      201

```

```

INVENTORY_ID PRODUCT_ID
-----

```

```

PRODUCT_NAME
-----

```

```

STORE_ID
-----

```

```

STORE_NAME
-----

```

```

SELLER_ID
-----

```

```

SELLER_NAME
-----

```

```

-----
QUANTITY_AVAILABLE    PRICE LAST_UPDA
-----

```

```

Classic Eyeglasses

```

```

INVENTORY_ID PRODUCT_ID
-----

```

```

PRODUCT_NAME
-----

```

```

STORE_ID
-----

```

```

STORE_NAME
-----

```

```

SELLER_ID
-----

```

```

SELLER_NAME
-----

```

9.2 Available and Unavailable Product count

```

SQL> SELECT
  2     CASE
  3         WHEN Quantity_Available > 0 THEN 'Available'
  4         ELSE 'Unavailable'
  5     END AS Stock_Status,
  6     COUNT(*) AS Total_Products
  7 FROM Inventory
  8 GROUP BY
  9     CASE
 10         WHEN Quantity_Available > 0 THEN 'Available'
 11         ELSE 'Unavailable'
 12     END;

STOCK_STATU TOTAL_PRODUCTS
-----
Available          5

```

```

STOCK_STATU TOTAL_PRODUCTS
-----
Unavailable         0

```

9.3 Total Stock by Seller

```

SQL> SELECT
  2     se.Seller_ID,
  3     se.Seller_Name,
  4     SUM(i.Quantity_Available) AS Total_Stock
  5 FROM Seller se
  6 JOIN Inventory i
  7     ON se.Seller_ID = i.Seller_ID
  8 GROUP BY
  9     se.Seller_ID,
 10     se.Seller_Name
 11 ORDER BY
 12     se.Seller_ID;

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      101
ABC Traders
      50

      102
Vision Opticals
      30

SELLER_ID
-----
SELLER_NAME
-----

```

```

-----
TOTAL_STOCK
-----

      103
Bright Eyewear
      45

      104
Clear Vision

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      25

      105
Eye Care Traders
      20

```

9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products.

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     se.Seller_ID,
  5     se.Seller_Name,
  6     i.Quantity_Available
  7 FROM Product p
  8 JOIN Inventory i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller se
 11     ON i.Seller_ID = se.Seller_ID
 12 WHERE i.Quantity_Available = 0;

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_ID
-----
SELLER_NAME
-----
QUANTITY_AVAILABLE
-----
      205
Blue Light Glasses
      105

PRODUCT_ID
-----

```

PRODUCT_NAME
SELLER_ID
SELLER_NAME
QUANTITY_AVAILABLE
Eye Care Traders
0